

Calculus 1: Homework Assignment #1

Math 101 - Section 1
Due date: Sunday 26/6/2022

Kuwait University
Department of Mathematics

Sections 1.2, 1.4-1.5

Question 1. For each of the following functions, find (1) $\mathcal{D}_f$, (2) $f^{-1}(x)$ and (3) $\mathcal{R}_f$:

(a) $f(x) = \ln(2x + 3)$.

(b) $f(x) = \ln(2e^x - 1)$.

(c) $f(x) = 4 + e^{x-3}$.

(d) $f(x) = \frac{x-1}{x+1}$

Question 2. Show that

(a) $\cos(\sin^{-1}(x)) = \sqrt{1-x^2}$.

(b) $\sin(\cos^{-1}(x)) = \sqrt{1-x^2}$

(c) $\sin(\tan^{-1}(x)) = \frac{x}{\sqrt{1+x^2}}$

Question 3. Find the exact value of

(a) $e^{2 \ln x}$

(b) $\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)$

(c) $\sin\left(\cos^{-1}\left(\frac{1}{2}\right)\right)$

(d) $\sin\left(\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

Sections 2.2, 2.3

Question 1. Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow -2} \frac{3x^2 + 5x - 2}{x - 2}$.

(b) $\lim_{x \rightarrow -2} \frac{3x^2 + 5x - 2}{x^2 - 3x - 10}$.

(c) $\lim_{t \rightarrow 2} \frac{t^2 - 4}{t^3 - 8}$.

(d) $\lim_{t \rightarrow 2} \frac{t^4 - 4}{t^2 - 4}$.

(e) $\lim_{x \rightarrow 1} \cos^{-1}\left(\frac{x-1}{x^2+1}\right)$.

(f) $\lim_{x \rightarrow 0} \frac{4^x - 1}{2^x - 1}$.

(g) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^x - 1}$.

(h) $\lim_{h \rightarrow 0} \frac{(h-4)^2 - 16}{h}$.

Question 2. Evaluate the following limits:

(a) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x^2 + x} \right)$

(b) $\lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{4x - x^2}$

(c) $\lim_{x \rightarrow 2} \frac{\sqrt{4x + 1} - 3}{x - 2}$

Question 3. Evaluate the following limits:

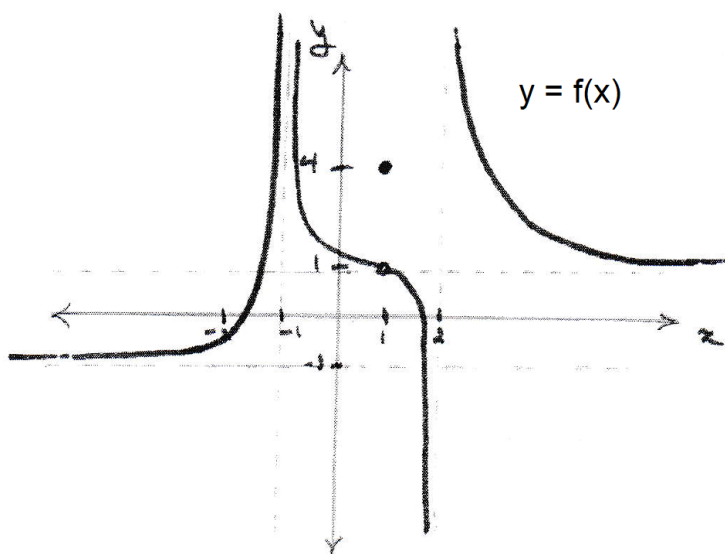
(a) $\lim_{x \rightarrow 0} x^2 \cos \left(\frac{1}{x} \right)$

(b) $\lim_{x \rightarrow 0} x e^{\sin(\frac{1}{x})}$

(c) $\lim_{x \rightarrow 1} (x - 1)^4 \cos \left(\frac{2}{x - 1} \right)$

(d) $\lim_{x \rightarrow 1} (x - 1)^2 \sin \left(\frac{\pi}{x - 1} \right)$

Question 4. The graph of the function $f(x)$ is shown below. Use the graph to evaluate the following limits:



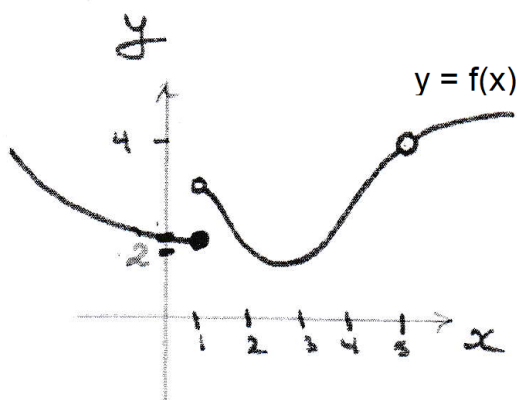
(a) $\lim_{x \rightarrow 1} f(x) =$

(b) $\lim_{x \rightarrow 2^-} f(x) =$

(c) $\lim_{x \rightarrow 2^+} f(x) =$

(d) $\lim_{x \rightarrow -1} f(x) =$

Question 5. The graph of the function $f(x)$ is shown below. Use the graph to evaluate the following limits:



(a) $\lim_{x \rightarrow 1^+} f(x) =$

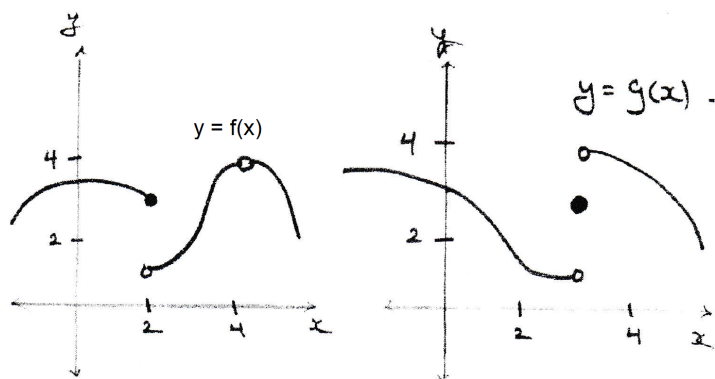
(b) $\lim_{x \rightarrow 1^-} f(x) =$

(c) $\lim_{x \rightarrow 1} f(x) =$

(d) $\lim_{x \rightarrow 5^+} f(x) =$

(e) $\lim_{x \rightarrow 5} f(x) =$

Question 6. The graph of the function $f(x)$ is shown below. Use the graph to evaluate the following limits:



(a) $\lim_{x \rightarrow 4} f(x) =$

(b) $\lim_{x \rightarrow 3} g(x) =$

(c) $\lim_{x \rightarrow 3^+} f(x)g(x) =$

(d) $\lim_{x \rightarrow 3^-} f(x)g(x) =$

(e) $\lim_{x \rightarrow 3} f(x)g(x) =$

Question 7. If

$$h(x) = \begin{cases} \sqrt{x-4}, & \text{if } x > 4, \\ 8-2x, & \text{if } x \leq 4, \end{cases}$$

evaluate the followings:

(a) $f(4) =$

(b) $\lim_{x \rightarrow 4^-} f(x) =$

(c) $\lim_{x \rightarrow 4^+} f(x) =$

(d) $\lim_{x \rightarrow 4} f(x) =$

Question 8. If $f(x) = \frac{|x-1|}{x^2-1}$, evaluate the followings:

(a) $f(1) =$

(b) $\lim_{x \rightarrow 1^-} f(x) =$

(c) $\lim_{x \rightarrow 1^+} f(x) =$

(d) $\lim_{x \rightarrow 1} f(x) =$

Homework #2 - Part 1

Math 101 - Section 1
Due date: Sunday 3/7/2022

Kuwait University
Department of Mathematics

Sections 2.3, 2.6

Question 1. Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2})$

(b) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - x + 1}}{x - 1}$

(c) $\lim_{x \rightarrow \infty} \frac{2e^x}{e^x - 1}$

(d) $\lim_{x \rightarrow -\infty} \sqrt{\frac{3x^2 - 1}{2x^2 + 3}}$

Question 2. Evaluate the following limits, if they exist.

(a) $\lim_{x \rightarrow 1} f(x), \quad \text{if } 1 \leq \frac{f(x)}{(x+1)^2} \leq \frac{x^3 - 1}{3(x-1)}$

(b) $\lim_{x \rightarrow 4} f(x), \quad \text{if } 4x - 9 \leq f(x) + x \leq x^2 - 4x + 7, \quad x \in \mathbb{R}$

(c) $\lim_{x \rightarrow 0} \frac{x}{2 + \sin\left(\frac{1}{x}\right)}$

(d) $\lim_{x \rightarrow \infty} \frac{x \sin(x)}{x^2 + 1}$

Question 3. Find the horizontal and vertical asymptotes, if any of them exists.

(a) $f(x) = \frac{|x|(2x^2 + 3)}{x^3 + 8}$

(b) $f(x) = \frac{(x^2 - 4)\sqrt{x^2 + 6}}{x^3 + x^2 - 6x}$

(c) $f(x) = \frac{x|x|}{x^2 - x}$

(d) $f(x) = \frac{\sqrt{x^2 + 1}}{x - 3}$

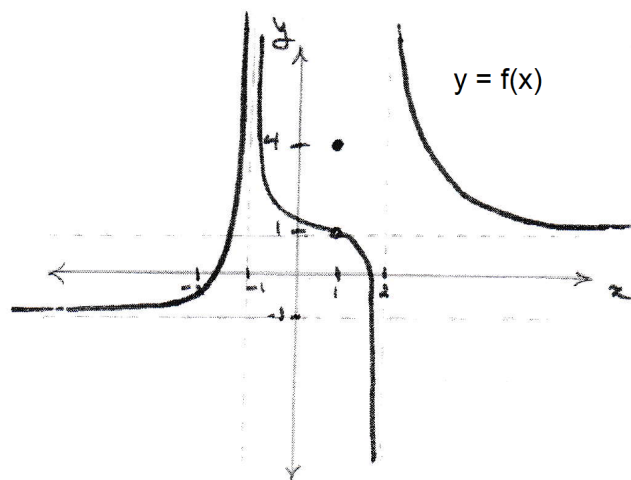
(e) $f(x) = \frac{2x|x-1|}{x^2 - 1}$

(f) $f(x) = \frac{x - 4}{|x| - 4}$

(g) $f(x) = \frac{3x^2|x-2|}{x^3 - 8}$

(h) $f(x) = \frac{x^2 - 4x + 4}{x^2 - 4}$

Question 4. Evaluate the following limits



(a) $\lim_{x \rightarrow -1} f(x) =$

(b) $\lim_{x \rightarrow 2} f(x) =$

(c) $\lim_{x \rightarrow -\infty} f(x) =$

(d) $\lim_{x \rightarrow \infty} f(x) =$

Homework #2 - Part 2

Math 101 - Section 1
Due date: Thursday 7/7/2022

Kuwait University
Department of Mathematics

Sections 2.4, 2.5

Question 1. Find all points of discontinuity $f(x)$ and classify each discontinuity as removable, jump or infinite.

(a) $f(x) = \frac{\sqrt{9-x^2}}{x^2-16}$.

(b) $f(x) = \frac{(x-2)|x-1|}{(x-1)(x^2-4)}$.

(c) $f(x) = \begin{cases} \frac{x^2+x-6}{x+3} & , \quad x < 0, \\ \frac{x+1}{(x-1)^2} & , \quad x \geq 0. \end{cases}$

(d) $f(x) = \begin{cases} \frac{2x+1}{x+3} & , \quad x < 0, \\ \frac{x^2-1}{x^2+x-2} & , \quad x \geq 0. \end{cases}$

(e) $f(x) = \begin{cases} x^2 - \sqrt{8-x} & , \quad x < -1, \\ -2 & , \quad x = -1, \\ \frac{x^2-3x+2}{x|x-2|} & , \quad x > -1. \end{cases}$

(f) $f(x) = \begin{cases} (2x+1)\cos(x) & , \quad x \leq 0, \\ \frac{1-\sqrt{x+1}}{x} & , \quad x > 0. \end{cases}$

(g) $f(x) = \begin{cases} \frac{x^2-x}{x^3} & , \quad x < -1, \\ 2 & , \quad -1 \leq x < 1, \\ 3 & , \quad x = 1, \\ x+1 & , \quad 1 < x \leq 2, \\ \frac{x^3-1}{x^2-3x+2} & , \quad x > 2. \end{cases}$

(h) $f(x) = \begin{cases} \frac{x-5}{x^2-6x+5} & , \quad x < 0, \\ \frac{x^2+x}{x^3-x} & , \quad 0 < x < 1, \\ \frac{-2|x-2|}{x^2-3x+2} & , \quad x > 1. \end{cases}$

Question 2. Use the $\varepsilon - \delta$ definition of the limit to prove the following limits:

(a) $\lim_{x \rightarrow 1} (-2x + 4) = 2$.

(b) $\lim_{x \rightarrow 1} (1 - 3x) = -2$.

Question 3. Find the values of A and B that make $f(x)$ continuous in $(-\infty, \infty)$.

$$f(x) = \begin{cases} \frac{2 \sin(x-1)}{x^2-1} & , \quad x > 1, \\ A + B \sin(\pi x/2) & , \quad x = 1, \\ 2A + \frac{x^3-1}{x-1} & , \quad x < 1. \end{cases}$$

Question 4. Find $A > 0$ that makes $f(x)$ continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{3 \tan(Ax)}{x} & , \quad x > 0, \\ 6A^2 + 5x & , \quad x \leq 0. \end{cases}$$

Question 5. Find the values of A and B that make $f(x)$ continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{3 \tan(3x)}{2x} & , \quad x < 0, \\ A & , \quad x = 0, \\ B \cos(x) & , \quad x > 0. \end{cases}$$

Question 6. State the Intermediate Value Theorem.

Question 7. Prove that the graphs of $f(x) = 2 + x \sin(x)$ and $g(x) = x^2$ intersect at least once.

Question 8. Prove that the equation $3^x + x - 2 = 0$ has at least one real solution.

Question 9. Prove that the equation $\sin(x) + x - 1 = 0$ has at least one real solution.

Question 10. Prove that the function $f(x) = x - \ln(x^3)$ has at least one real root in $[1, e]$.

Question 11. Prove that there is at least one $c \in (a, b)$ such that $f(c) = g(c)$, where $f(x)$ and $g(x)$ are continuous on $[a, b]$ and satisfy

$$f(a) > g(a), \quad f(b) < g(b).$$

Homework #3

Math 101 - Section 1
Due date: Monday 25/7/2022

Kuwait University
Department of Mathematics

Sections 2.7 - 2.8

Question 1. At the given points, find (a) an equation of the tangent line, and (b) an equation of the normal line:

(a) $y = x^3 + x - 1$, $(1, 1)$.

(b) $y = \sqrt{x}$, $(4, 2)$.

(c) $y = (1 - x^2)e^x$, $(0, 1)$.

(d) $y = \frac{2x + 1}{x + 2}$, $(1, 1)$.

Question 2. If an equation of the tangent line to the curve $y = f(x)$ at the point where $x = 2$ is $y = 4x - 5$, find $f(2)$ and $f'(2)$.

Question 3. If the tangent line to $y = f(x)$ at $(4, 3)$ passes through the point $(0, 2)$, find $f(4)$ and $f'(4)$.

Question 4. Evaluate the following limits:

(a) $\lim_{x \rightarrow 1} \frac{x^{2017} - 1}{x - 1}$.

(b) $\lim_{x \rightarrow 1} \frac{x^3 - 2x + 1}{x - 1}$.

(c) $\lim_{x \rightarrow 1} \frac{x^4 + x - 2}{x - 1}$.

(d) $\lim_{x \rightarrow 16} \frac{\sqrt[4]{x} - 2}{x - 16}$.

(e) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x - 1}$.

(f) $\lim_{x \rightarrow 3} \frac{3^x - 27}{x - 3}$.

(g) $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.

(h) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(x) - 1}{x - \frac{\pi}{2}}$.

(i) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x - \frac{\pi}{2}}$.

(j) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan(x) - 1}{x - \frac{\pi}{4}}$.

Question 5. Determine whether $f'(0)$ exists for the following functions:

$$\begin{aligned} \text{(a)} \quad f(x) &= \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & 0 = 0 \end{cases} \\ \text{(b)} \quad f(x) &= \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & 0 = 0 \end{cases} \end{aligned}$$

Sections 3.1-3.2

Question 6. Differentiate the following functions:

$$\begin{aligned} \text{(a)} \quad f(x) &= 4^2 - 3. \\ \text{(b)} \quad f(x) &= x^2 - 3x + 1. \\ \text{(c)} \quad f(x) &= e^2. \\ \text{(d)} \quad f(x) &= 3x^2 - e^x. \\ \text{(e)} \quad f(x) &= 2^{100}. \\ \text{(f)} \quad f(x) &= 2^e. \\ \text{(g)} \quad f(x) &= \sqrt[3]{\pi}. \\ \text{(h)} \quad f(x) &= \sqrt[3]{x} - \frac{2}{\sqrt{x}}. \\ \text{(i)} \quad f(x) &= \cos(\pi). \\ \text{(j)} \quad f(x) &= x^2(1 - x^2). \\ \text{(k)} \quad f(x) &= \frac{x^2}{1 - e^x}. \end{aligned}$$

Question 7. Find the points of the given curve where the tangent is (1) horizontal, and (2) vertical:

$$\begin{aligned} \text{(a)} \quad y &= 2x^3 + 3x^2 - 12x + 1. \\ \text{(b)} \quad y &= e^x - 2x. \end{aligned}$$

Question 8. Show that the curve $y = 2e^x + 3x + 5x^3$ has no tangent line with slope 2.

Question 9. Find an equation of the tangent line to the curve $y = x\sqrt{x}$ that is parallel to the line $y = 1 + 3x$.

Question 10. Find an equation of the normal line to the curve $y = x^2 - 5x + 4$ that is parallel to the line $x - 3y = 5$.

Homework #4

Math 101 - Section 1
Due date: Sunday 31/7/2022

Kuwait University
Department of Mathematics

Sections 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.9, 3.10, 3.11

Question 1. Evaluate the following limits:

(a) $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 + x - 6}.$

(b) $\lim_{x \rightarrow 0} \frac{\tan(x) \sin(2x)}{x^2}.$

(c) $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{\tan(x)}.$

(d) $\lim_{x \rightarrow \infty} \frac{\sinh(x)}{e^x}.$

Question 2. Show that the left-hand-side is equal to the right-hand-side:

(a) $\frac{d}{dx} [\sin^{-1}(2x)] = \frac{2}{\sqrt{1-4x^2}}.$

(b) $\frac{d}{dx} \sinh(x) = \cosh(x).$

(c) $\frac{d}{dx} [\sec^{-1}(x)] = \frac{1}{x\sqrt{x^2-1}}.$

Question 3. Differentiate the followings:

(a) Find $\frac{dy}{dx}$ if $y = x^{\sin(x)}.$

(b) Find $\frac{dy}{dx}$ if $y = (3x^2 - 2x + 1)^{\tan(x)}.$

(c) Find $\frac{dy}{dx}$ if $y = (x^2 + 1)^{\ln(x)}.$

(d) Find $\frac{dy}{dx}$ if $x^2 e^y + y \sinh(x) = 4.$

(e) Find $\frac{dy}{dx}$ if $y = 2^x \cosh(x) + [\sec^{-1}(x)]^5.$

(f) Find $\frac{dy}{dx}$ if $e^{xy} + 2y - 3x = \sin(y).$

(g) Find $\frac{dy}{dx}$ if $y = x^2 y^2 + x \sin(y) = y.$

Question 4. Find an equation of the tangent and normal lines to the following curves at the given points:

(a) $\sin(x+y) = (x+2)(y+2)$ at $P(2, -2).$

(b) $y^3 + x^2 y + x^2 - 3y^2 = 0$ at $P(0, 3).$

(c) $2y + x^2 y + x \sin(x^2) = e^{xy} + 1$ at $P(0, 1).$

(d) $y = x^{\ln(x)}$ at $P(1, 1).$

(e) $y = \sin(xy^2)$ at $P(\frac{\pi}{2}, 1).$

Question 5. Given the function $f(x)$ use Linear Approximation to estimate its value at nearby points.

- (a) $f(x) = e^x \cos(x)$, estimate $f(0.1)$.
- (b) $f(x) = \ln(x)$, estimate $\ln(1.1)$.
- (c) $f(x) = \sqrt{5 + x^2}$, estimate $f(1.95)$.
- (d) $f(x) = e^x$, estimate $e^{0.01}$.
- (e) $f(x) = \sqrt[3]{x}$, estimate $\sqrt[3]{26}$.
- (f) $f(x) = \sqrt{4x + 1}$, estimate $f(2.03)$.

Question 6. Estimate the following real numbers using linear approximation of the proper function

- (a) $\sqrt[4]{16.08}$.
- (b) $\tan(44^\circ)$.
- (c) $\sec(59^\circ)$.
- (d) $\sqrt{15.9}$.

Question 7. Use Linear Approximation to show that

- (a) $\sqrt{2x + 1} \approx 1 + x$, where $a = 0$.
- (b) $\sin(x) \approx x$, where $a = 0$.
- (c) $\cos(x) \approx 1$, where $a = 0$.
- (d) $e^x \approx 1 + x$, where $a = 0$.
- (e) $\ln(1 + x) \approx x$, where $a = 0$.

 **Question 8.** The followings are word problems for rectangular shapes. Use related rates to solve them.

- (a) Each side of a square is increasing at a rate of 6 cm/sec. At what rate is the area of the square increasing when the area is 25 cm^2 .
- (b) The length of a rectangle is increasing at a rate of 10 cm/sec while its width is decreasing at a rate of 4 cm/sec. When the length is 40 cm and the width is 15 cm, how fast is the area of the rectangle changing?

 **Question 9.** The followings are word problems for cylindrical shapes. Use related rates to solve them.

- (a) If a cylindrical tank of diameter 6m is being filled with water at a rate of $2 \text{ m}^3/\text{hr}$, how fast is the water level rising?

 **Question 10.** The followings are word problems for circular & spherical shapes. Use related rates to solve them.

- (a) The surface area of a sphere increases at a rate of $48\pi \text{ cm}^2/\text{sec}$. Find the rate of change of the volume of the sphere when its surface area is 36 cm^2 .
- (b) When a spherical ball made of steel is heated, its surface area S (in cm^2) after t hours is given by $S = \sqrt{t^2 + t - 4}$. Find the rate at which the radius of the ball is changing after 4 hours.
- (c) Oil spilled from a tanker spreads in a circle whose area increases at a constant rate of $9 \text{ km}^2/\text{hr}$. How fast is the radius of the spill increasing when its area is 16 km^2 ?

Question 11. The followings are word problems for triangular shapes. Use related rates to solve them.

- (a) A 5-meter ladder rests against a vertical wall. If the top of the ladder is sliding down the wall at a rate of 0.9 m/s, how fast is the bottom of the ladder sliding away from the wall when the top of the ladder is 4m from the ground?
- (b) A rocket, rising vertically, is tracked by a radar station that is on the ground 4 km from the launchpad of the rocket. How fast is the rocket rising when it is 3 km high and its distance from the radar station is increasing at a rate of 1500 km/hr?

Question 12. The followings are word problems for conic shapes. Use related rates to solve them.

- (a) A water tank has the shape of an inverted circular cone with base radius 2m and height 4m. If water is being pumped into the tank in such a way that the radius of the surface of the water, r , increases at a rate of 2 m/min, find the rate at which the volume of the water is changing when the radius of its surface is $r = 1$ m.

Question 13. The followings are word problems for a particle in a coordinate plane. Use related rates to solve them.

- (a) A particle is moving along the curve $y = \sqrt{1 + x^3}$ in the $x - y$ plane. If the y -coordinate is increasing at a rate of 4 cm/sec at $x = 2$, how fast is the x -coordinate changing?
- (b) A particle is moving along the curve $y = x^2 + 3x + 1$. As it reaches the point $(1, 5)$, the y -coordinate is decreasing at a rate of 2 cm/sec at $x = 2$. How fast is the x -coordinate of the particle changing at that instant?

Homework #5

Math 101 - Section 1
Due date: Thursday 11/8/2022

Kuwait University
Department of Mathematics

Sections 4.1, 4.2, 4.3, 4.5, 4.7

Question 1. Find the absolute maximum and minimum values of the following functions on the given intervals:

(a) $f(x) = 3x - 5x^{3/5}$ on $[-1, 32]$.

(b) $f(x) = xe^{-x}$ on $[0, \infty)$.

Question 2. For each function $f(x)$, sketch the graph of f :

(a) $f(x) = \frac{e^x}{x^2}$.

(b) $f(x) = \frac{3x^2}{x^2 - 4}$ and $f'(x) = \frac{-24x}{(x^2 - 4)^2}$.

(c) $f(x) = \left(\frac{x}{x+1}\right)^2$, $f'(x) = \frac{2x}{(x+1)^3}$ and $f''(x) = \frac{2(1-2x)}{(x+1)^4}$.

Question 3. State Rolle's Theorem.

Question 4. State the Mean Value Theorem.

Question 5. If $f(x)$ is continuous on $[a, b]$, and $f'(x) = m$ for every $x \in (a, b)$, use the Mean Value Theorem to show that $f(x) = mx + d$ for some real number d .

Question 6. If $2 < f'(x) < 4$ for all $x \in \mathbb{R}$, show that $10 < f(6) - f(1) < 20$.

Question 7. Show that if $0 < a < b$, then $e^b < e^a + b - a$.

Question 8. Use the Mean Value Theorem to show that for $x > 0$

$$\tan^{-1}(x) < x.$$

Question 9. Use the Mean Value Theorem to show that for $x > 0$

$$e^x > 1 + x.$$

Question 10. Let $f(x) = |x - 1|$.

(a) Show that there is no value $c \in (0, 3)$ such that $f'(c) = \frac{1}{3}[f(3) - f(0)]$.

(b) Explain why the result in part (a) does not contradict the Mean Value Theorem?

Question 11. Show that the following equations have at most one real solution:

(a) $2x + e^x = 0$.

(b) $x^3 + ax = 1, a > 0$.

Question 12. Prove that

$$2 \sin^{-1}(x) = \cos^{-1}(1 - 2x^2), \quad x \geq 0.$$

Question 13. The followings are optimization problems. Convert each of them to a direct mathematical problem that can be solved by finding the absolute extreme value of the objective function and the dimensions leading to it.

(a) Find the point on the curve $y = \sqrt{x}$ that is closest to the point $(2, 0)$.

(b) Find two nonnegative numbers x and y whose sum is 15 and $P = x^2y^3$ is a maximum.

(c) Find the maximum product of two positive x and y such that $2x + y = 100$.

(d) Find the dimensions of a rectangle with area 100 m^2 whose perimeter is as small as possible.

(e) If the sum of the radius and height of a cylinder is 3 m, find the dimensions of the cylinder that maximize its volume.

Homework #6

Math 101 - Section 1
Due date: Any time!

Kuwait University
Department of Mathematics

Sections 5.2, 5.3, 5.4, 5.5, 6.1

Question 1. Show that $F(x) = \frac{1}{2}(x^2 \tan^{-1} x - x + \tan^{-1} x)$ is an antiderivative of $f(x) = x \tan^{-1} x$.

Question 2. Compute the Riemann sum of the following functions:

(a) $f(x) = \sin x$, $x \in [0, \pi]$, $n = 3$ and the sample points are the right endpoints.

(b) $f(x) = 1 + x^3$, $x \in [1, 4]$, $n = 3$ and the sample points are the left endpoints.

Question 3. Show that $2 \leq \int_0^1 (3 - \cos^5(7x + 1)) dx \leq 4$.

Question 4. Given $\int_{-2}^1 f(x) dx = 2$ and $\int_3^1 f(x) dx = 6$, find $\int_3^{-2} f(x) dx$.

Question 5. Given $\int_1^5 f(x) dx = 7$, find the value of $\int_1^5 (3f(x) + x) dx$.

Question 6. Evaluate the following definite integrals

(a) $\int_{-1}^1 (x^2 + \cos x) \tan x dx$

(b) $\int_{-1}^1 x^2 \tan x dx$

Question 7. Evaluate the following integrals

(a) $\int \frac{dx}{x (\ln x)^5}$

(b) $\int \frac{1}{\sqrt{x} (\sqrt{x} + 1)^2} dx$.

(c) $\int \sin^3(x) \cos(x) dx$.

(d) $\int 2 \cos(2x) \sqrt{1 + \sin(2x)} dx$

(e) $\int \cos(x) \sin[\sin(x)] dx$.

(f) $\int \frac{x^2}{\sqrt{x^2 + 1}} dx$.

Question 8. Evaluate the following integrals

- (a) $\int_0^{\pi/4} \frac{2 \sec^2 x}{1 + \tan x} dx$
- (b) $\int_1^2 \frac{x + 2}{x^2 + 4x + 4} dx$
- (c) $\int_e^{e^2} \frac{1}{x \ln x} dx$
- (d) $\int_1^e \frac{\ln(x^2)}{x} dx$

Question 9. Evaluate the following definite integrals

- (a) $\int_0^2 (x + \sqrt{4 - x^2}) dx$
- (b) $\int_{-1}^1 \sqrt{1 - x^2} dx$

Question 10. Given the following functions, find the equation of the tangent and normal lines at the given points:

- (a) $f(x) = \int_{-1}^{x^3} \frac{\ln(t + 2)}{t^2 + 1} dt$, at $x = -1$.
- (b) $h(x) = \int_{\sqrt{x}}^{x^2} \ln(1 + 2t) dt$ at $x = 1$.

Question 11. Find the absolute minimum value, if any, of the function $f(x) = \int_1^{x^2+1} \frac{t^2}{\sqrt{t^2 + 1}} dt$.

Question 12. On what interval is the curve $y = \int_1^{e^x} \ln(t) dt$ concave upward.

Question 13. Find the inflection points, if any, of the curve

$$y = \int_0^x (1 - t)e^t dt.$$

Question 14. Sketch the region enclosed by each the following curves and find its area:

- (a) $f(x) = \sqrt{x - 1}$ and $g(x) = x - 1$.
- (b) $y = \sqrt{x} + 1$ and $y = x^3 + 1$.
- (c) $y = x - 2$ and $y = x^2 - 4$.